

ABL Customer Statement Intelligence Suite

Final Technology Stack

Component	Selected Technology	Reason
Frontend	React.js + Tailwind CSS	React provides reusable UI components; Tailwind provides ready-made utility classes for faster styling.
Build Tool	Vite	Lightweight tool used to run and build the React frontend.
Frontend Deployment	Vercel	Easy frontend deployment with Git-based builds and a free tier.
Backend	FastAPI	Lightweight Python backend that works well with APIs and AI/ML libraries.
Backend Deployment	Heroku	Straightforward FastAPI deployment and available student benefits help avoid hosting cost.
LLM	Google Gemini API - Gemini 3.7 Flash	Fast and capable model suitable for the project's AI question-answering tasks.
Embedding Model	all-MiniLM-L6-v2 (384D)	Free and lightweight model for semantic similarity search.
Database	PostgreSQL	Reliable relational database suitable for structured application data.
Vector Database	PostgreSQL + pgvector	Provides vector similarity search inside PostgreSQL without requiring another database.

Architecture: React frontend on Vercel communicates with the FastAPI backend on Heroku. Gemini handles language generation; all-MiniLM-L6-v2 creates embeddings; PostgreSQL stores structured data; and pgvector handles vector similarity search.

Cost: The stack is selected around free/open-source technologies and free or student-tier services for development and demonstration.